LESSON PLAN

EMI/EMC (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	45
Total No. of Lesson Plans	46	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Architecture and control laws of a typical FBW system.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Architecture and control laws of a typical FBW system. Advantages and disadvantages of an FBW system.
5. Specific Objectives	14.2.1.1 Trainee will be able to explain the functions of AFCS.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of FBW block diagrams and control law definitions from IB; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the AFCS?
(b) Define the term 'fly-by-wire' in the context of modern aircraft.
(c) List three traditional mechanical control linkages used in older aircraft.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• FBW System Architecture Overview • Sensors and Signal Processing • Flight Control Computers (FCC) • Actuator Control Electronics • Control Law Logic Implementation	L & Ds	PPT, IB	15 min
(b)	• Gain Scheduling Principles • Stability Augmentation Systems • Protection Envelopes • Direct Law vs Normal Law • Degraded Mode Operations	L & Ds	PPT, PDF	20 min
(c)	• Advantages of FBW Systems • Disadvantages and Risks • Redundancy Requirements • Cybersecurity Considerations • Comparison with Mechanical Systems	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how a control law modifies pilot input to ensure aircraft stability.
(b) Identify two major advantages and two major disadvantages of implementing an FBW system.
(c) Describe the role of the Flight Control Computer in processing sensor data.

12. Home Assignment and Follow Up	Prepare a brief note comparing the pitch control interceptor logic in a typical FBW system versus a conventional mechanical system, referencing concepts introduced in LP 46.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Explanation of pitch and roll control interceptors.	Refer to the master syllabus reference list	LP 46

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min